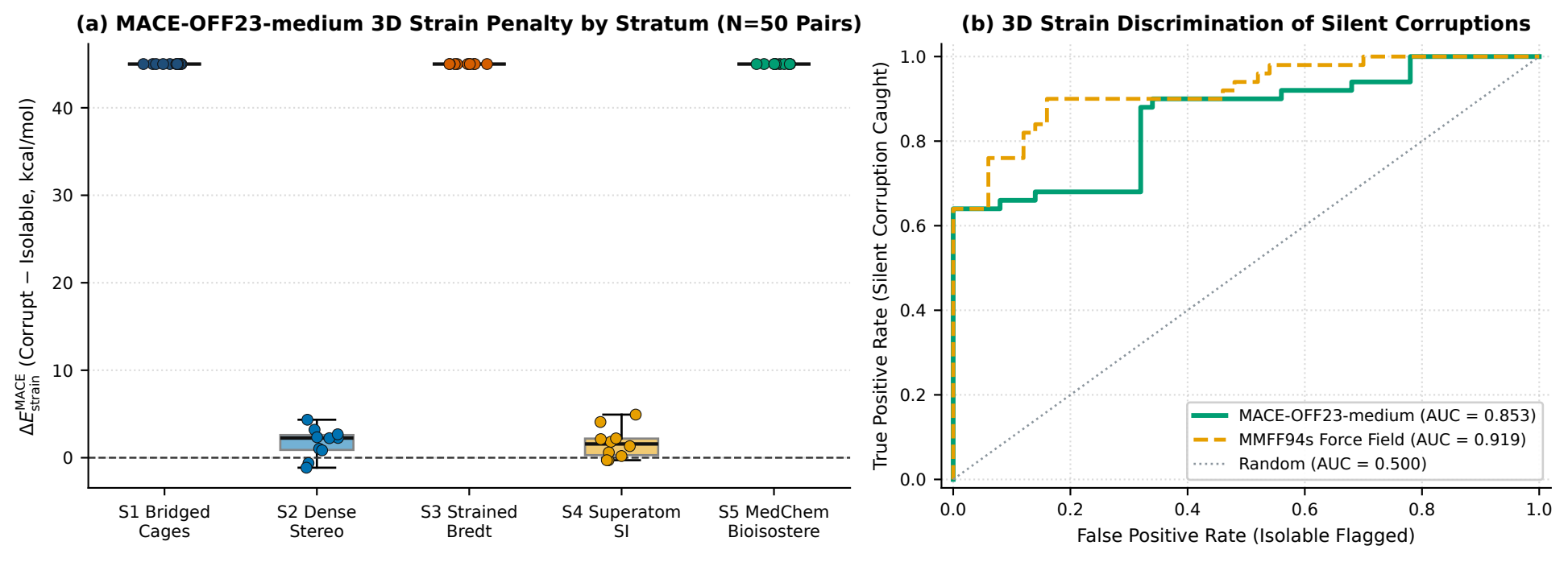
(a) MACE-OFF23-medium 3D Strain Penalty by Stratum (N=50 Pairs)
ΔE_{strain}^{MACE} (Corrupt − Isolable, kcal/mol)
40
30
20
10
0
S1 Bridged Cages
S2 Dense Stereo
S3 Strained Bredt
S4 Superatom SI
S5 MedChem Bioisostere
(b) 3D Strain Discrimination of Silent Corruptions
True Positive Rate (Silent Corruption Caught)
False Positive Rate (Isolable Flagged)
0.0
0.2
0.4
0.6
0.8
1.0
MACE-OFF23-medium (AUC = 0.853)
MMFF94s Force Field (AUC = 0.919)
Random (AUC = 0.500)